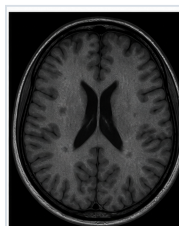


MRI contrasts and data representations used by AI for MS

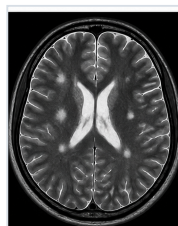
(a) Registered brain contrasts and specialized MS views

Same synthetic brain case; registered contrasts



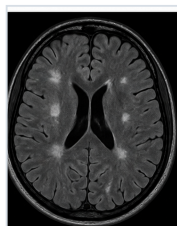
T1-weighted

anatomy /
atrophy



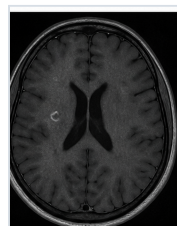
T2-weighted

water-sensitive
signal



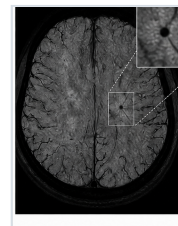
FLAIR

WM lesion
conspicuity



Post-contrast T1

enhancing
activity



Susceptibility

susceptibility-
related cues

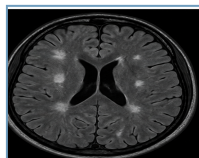


Cervical cord

cord lesion
assessment

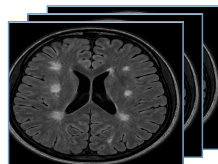
Specialized acquisitions

(b) Network input representations



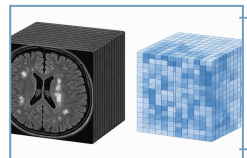
2-D

single slice



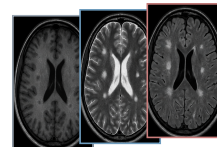
2.5-D

adjacent-slice channels



3-D

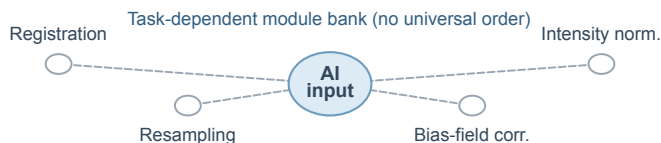
volumetric context



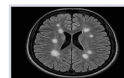
Multisequence

registered contrast channels

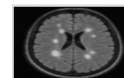
(c) Prespecified preprocessing choices and domain variables



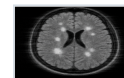
Same synthetic anatomy across image domains



vendor / field



protocol



reconstruction